LLM-Driven Multi-Agent Architectures for Intelligent Self-Organizing Networks

Adnan Qayyum, Abdullatif Albaseer, Junaid Qadir, Ala Al-Fuqaha, and Mohamed Abdallah

Abstract

Managing the growing complexity of Self-Organizing Networks (SONs) in next-generation communication systems requires agile, real-time strategies that can adapt to multidimensional and highly dynamic conditions. Traditional SON management rooted in centralized, rule-based, and static models, struggles to meet these evolving requirements. Recent advances in multi-agent systems (MAS) and Large Language Models (LLMs) enable the design of intelligent and context-aware frameworks that span multiple operational layers. In this paper, we introduce LaMA-SON, an LLM-driven MAS for intelligent SON management. Specifically, LaMA-SON incorporates specialized agents to handle three critical operational functions: traffic management, quality of service (QoS) optimization, and security threat detection. We perform a proof-of-concept evaluation using a real-world network traffic classification dataset, where traffic, security, and QoS optimization agents make decisions based on role-specific features and structured prompts. Our experimental results demonstrate that LaMA-SON achieves higher accuracy and recall while preserving balanced precision-recall trade-offs and outperforms standalone LLM baselines, highlighting the benefits of multi-agent collaboration. Finally, we highlight various open research challenges that require further investigation to fully realize the potential of LLM-based MAS frameworks in network operations management.

Introduction

Advancements in wireless communication—particularly the rollout of 5G and the development of 6G—are reshaping how networks are managed. Modern networks now support a broad ecosystem of applications, including smart cities, metaverses, autonomous vehicles, and intelligent healthcare. These emerging services require ultra-low latency, high adaptability, and resilience [1]. While effective in earlier generations, traditional centralized and rule-based network management approaches cannot scale to today's highly dynamic environments. To meet these demands, Self-Organizing Networks (SONs) have been introduced to provide autonomous capabilities such as self-configuration, self-optimization, and self-healing. Although many SON implementations incorporate dynamic models, these are often limited to narrow tasks and lack the capacity for holistic and context-aware reasoning across diverse operational scenarios. In particular, they struggle to generalize across tasks or coordinate multiple concurrent objectives in real-time. As networks evolve towards 6G with intelligent edge processing and integrated sensing-communication [2], there is a critical need for intelligent, context-aware, and adaptable network management frameworks.

In this regard, the recent advancements in Large Language Models (LLMs) offer a new frontier for empowering edge intelligence in SONs. Unlike traditional machine and deep learning models, LLMs exhibit powerful generalization and contextual reasoning capabilities. They can process complex structured and unstructured network data, including configuration files, telemetry data, and traffic logs [3]. Their rise coincides with the growing need for edge intelligence in wireless systems, where deploying Artificial Intelligence (AI) capabilities directly at the network edge, particularly within base stations (BSs), has become increasingly crucial. This paradigm shift enables real-time responsiveness to traffic dynamics, interference, user mobility, and security threats, while mitigating the latency and scalability limitations of centralized architectures [4]. However, achieving true autonomy at the edge requires more than decentralized inference, and it demands reasoning and coordination mechanisms that can operate contextually and collaboratively.

Despite their potential, LLMs alone are insufficient for optimizing network operations at scale. They require structured coordination and task-specific expertise to function effectively in complex, real-time network environments. A promising solution is the integration of LLMs within intelligent Multi-Agent Systems (MAS), which is well-established for task decomposition, collaborative intelligence, real-time situational awareness, and autonomous task execution [5]. This synergy, which integrates deep LLM-based reasoning, agent-level adaptability, and MAS coordination, represents a significant step toward achieving intelligent, adaptive, and resource-efficient network management.

The potential of integrating LLMs has been demonstrated for various networking applications such as network slicing [8], zero-touch automation [9], intent-based networking [10], and network planning [11]. Also, domain-specialized foundation models like TelecomGPT [12] have been

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shown to outperform general-purpose LLMs in tasks such as protocol parsing and fault analysis. Most of the existing studies are generally focused on highlighting the potential of LLMs in different network applications without considering the complex network management task. This resulted in one single article that proposes the use of LLM agents in 6G communication with a case study on semantic communication [6]. Table 1 compares this paper with existing articles that have a similar focus. It is evident from the table that previous studies remain fragmented and do not primarily focus on empowering edge intelligence within SONs through LLM agents. This work is inspired by the framework proposed by Kim et al. [7], and we extend the idea of collaborative multi-agent systems to dynamic operations management in SON. While this is an inspiring foundation, their framework has several differences: it relies on proprietary models (e.g., Gemini, GPT-4), does not implement feature-role alignment, and is evaluated on medical QA tasks with no networking context.

Fig. 1 presents a high-level overview of our architecture, where multiple LLM-based agents are deployed at BSs to support real-time decentralized adaptive decision-making at the network edge. These agents collectively realize the vision of edge intelligence in SONs and embody the core principles of self-configuration, self-optimization, and self-healing. Within each BS, agents collaborate locally to address network issues while exchanging critical information globally through shared cloud memory. This architecture exemplifies decentralized intelligence, where each BS autonomously processes data, executes decisions, and adapts to local conditions without depending on centralized control. Meanwhile, inter-BS communication and synchronization ensure consistency across the broader network. In addition, each specialized agent will be equipped with role-specific tools and reasoning capabilities tailored to distinct aspects of network requests, enabling fine-grained and context-aware management. Our proposed **L**arge language model-based **M**ulti-**A**gent **S**ystem for **S**elf-**O**rganizing Networks (LaMA-SON) framework extends beyond traditional static and rule-based methods by enabling dynamic, collaborative adaptation to real-time network conditions through LLM-based agents. To the best of our knowledge, this is the first to present the application of LLM-based multi-agent coordination tailored to SON operations. The following are the salient contributions of this paper.

1. We propose LaMA-SON, a multi-agent framework powered by open-source LLMs for intelligent network operations management in SONs.
2. We present a proof-of-concept evaluation of LaMA-SON using a real-world network traffic task in a proxy experimental setting, where agents operate on feature-role mappings aligned with domain-specific expertise to demonstrate multi-agent collaboration and inference.
3. We compare LaMA-SON against standalone LLMs and show that the proposed multi-agent architecture delivers more balanced and robust performance across evaluation metrics, thereby validating the benefits of agent specialization and dialogue-based refinement in complex decision-making scenarios.

LaMA-SON: LLM-Based Multi-Agents for Self-Organizing Networks

Our proposed framework leverages multiple LLM-based agents for collaborative SON management. As illustrated in Fig. 2, LaMA-SON incorporates multiple LLMs, each designated for a specific role. These LLMs are strategically deployed at BSs (Fig. 1) to enable localized decision-making and real-time adaptability. LaMA-SON leverages the generalization

Reference	Focus Area	Contributions	Case Study	LLMs	SONs	Complexity Checking	Multiple Roles for Agents	Multi-Agent Chat	Evaluation, Feedback, Review	Shared Cloud Memory	Distributed Coordination
Jiang et al. [6]	6G Communication	Multi-agent system for retrieval, planning, and evaluation	Semantic communication	GPT-3.5 (•)	×	×	✓	✓	✓	×	×
Kim et al. [7]	LLMs in Healthcare	Collaborative LLM agent framework for healthcare decision-making	Multi-modal question answering	Gemini (•), GPT-4 (•)	×	✓	✓	✓	✓	×	×
Long et al. [4]	LLMs for 6G	LLM-based architecture for network operation and optimization	Network health management	N/A	×	×	×	×	×	×	×
Dandoush et al. [8]	LLMs for Slicing	Vision for LLM agents in network slicing and orchestration	Network slicing	N/A	×	×	✓	✓	×	×	×
Ours	LLM Agents for SON	LLM-based multi-agent framework for self-configuration, self-healing, and self-optimization in SONs	Network traffic classification	LLaMA-3 (◦), LLaMA-3.2 (◦), Qwen2.5 (◦), Cogito (◦)	✓	✓	✓	✓	✓	✓	✓

TABLE 1. Comparison of our paper with related prior work. Closed-source models are indicated with (•) and open-source models with (◦).

capabilities of specialized LLMs, allowing agents to reason over complex and evolving network issues without requiring exhaustive feature engineering. Moreover, this integration enables LaMA-SON to interpret long-context network events, adapt to unforeseen issues, and facilitate inter-agent collaboration for multi-step decision-making. LaMA-SON envisions the development of small domain-specific foundation models [13], [12], which are tailored to unique

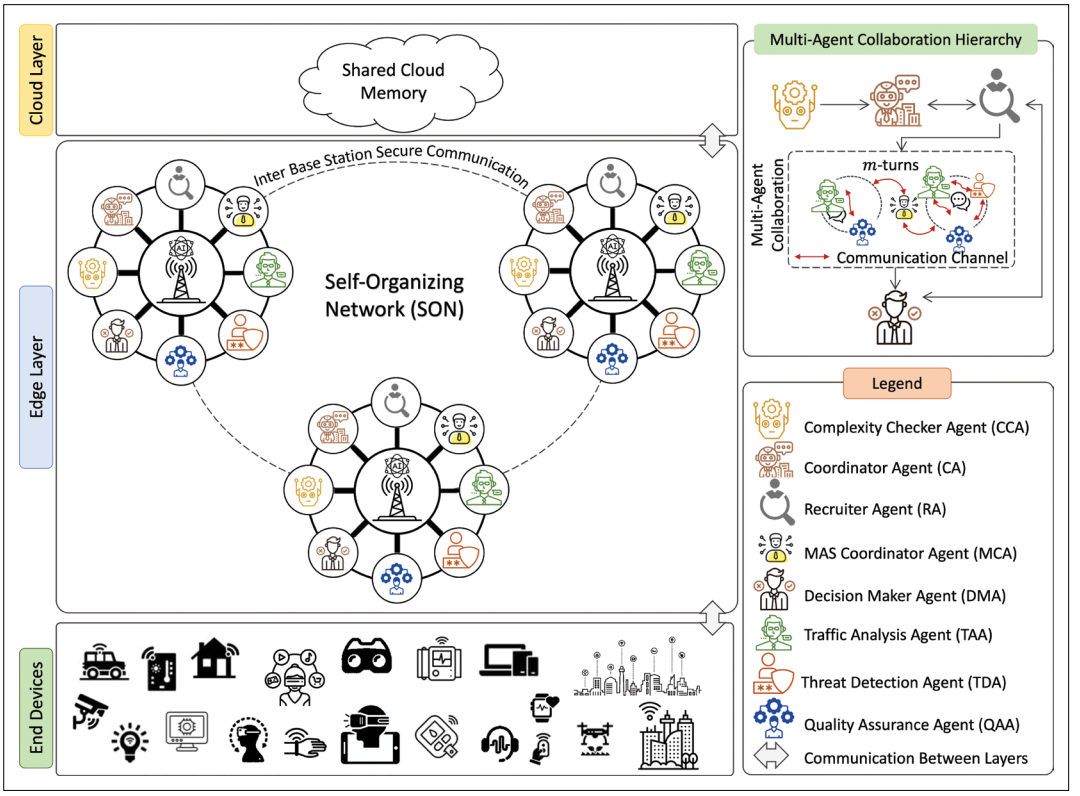


FIGURE 1. High-level illustration of empowering edge intelligence in SONs through multiple LLM-based agents collaborating locally at BSs for real-time, autonomous, and decentralized network management.

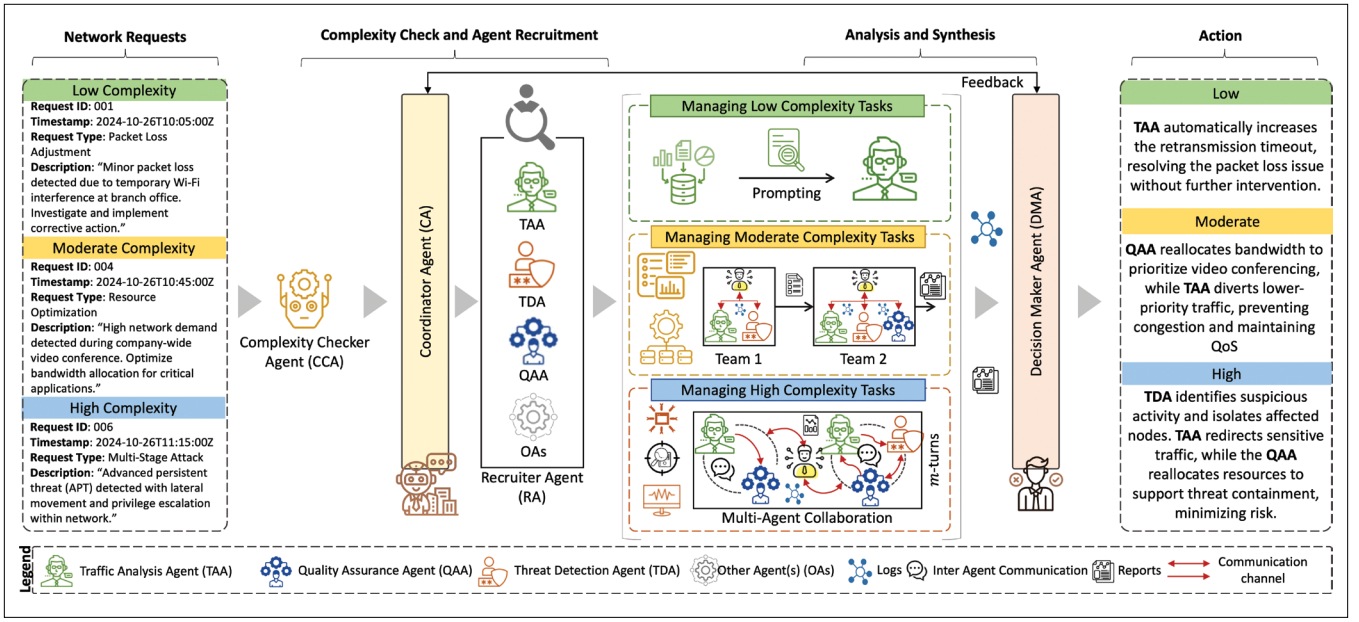


FIGURE 2. Proposed LaMA-SON framework is deployed at each BS in SONs for the dynamic management of network issues locally (Fig. 1). It employs an LLM-based multi-agent architecture to adaptively recruit domain-specific agents based on the assessed complexity of incoming network requests. For moderate and high-complexity cases, LaMA-SON enables collaborative decision-making among multiple agents to improve response accuracy and operational robustness. *Illustrated network requests and actions are representative examples.*

network operations, e.g., traffic management, security management, and QoS optimization. These specialized models would enhance the reasoning, predictive analytics, and response precision of their respective domains. In the following, we describe the operational scope of each LLM-based agent and its interaction logic in the LaMA-SON framework.

Complexity Checker Agent (CCA)

CCA is the initial reasoning component in the LaMA-SON framework, which is responsible for assessing the relative complexity of incoming network requests. Using predefined criteria and contextual analysis, it provides a lightweight yet critical triage function by classifying network issues into three discrete levels: *low, moderate*, or high complexity. This classification determines the depth and breadth of specialized agent involvement required for downstream processing. The CCA's role is lightweight, designed to support early-stage decision branching without incurring unnecessary computational overhead. This facilitates rapid agent recruitment and supports scalable deployment across distributed network environments.

- *Low Complexity*: Routine issues such as minor packet loss and simple routing issues can be handled independently by a single specialized agent.
- *Moderate Complexity*: Issues involving multiple interrelated factors—such as traffic spikes combined with low-level security alerts, require coordination between agents (e.g., traffic and security agents) to balance the load and maintain protection.
- *High Complexity*: Large-scale challenges such as system-wide failures, DDoS attacks, or multi-layer network optimization require extensive collaboration across multiple agents. For instance, mitigating DDoS-induced congestion may involve coordinating traffic, security, and QoS agents to restore stable service.

Coordinator Agent (CA)

CA acts as the central orchestrator in the LaMA-SON framework, managing inter-agent workflows and ensuring coherent task execution. It synthesizes telemetry data obtained from diverse sources, including Simple Network Management Protocol (SNMP), NetFlow (flow-level traffic summaries), and logs generated by intrusion detection systems (for security event tracking). The following are the key responsibilities of the network coordinator agent.

- *Task Dispatching*: Upon receiving the complexity level from the CCA, the CA delegates the input request to a dedicated agent responsible for recruiting domain-specialized agents.
- *Monitoring and Feedback Collection*: The CA collects feedback on task outcomes and agents' performance. This feedback is essential for refining future task assignments and improving the overall system's efficiency through a continuous feedback loop.
- *Decision-Making and Escalation*: In critical scenarios, the CA synthesizes the recommendations of collaborating agents, and if necessary, it can escalate decisions to supervisory agents or human operators.
- *Logging and Reporting*: The CA maintains a comprehensive activity log, tracking agent interactions, task resolutions, and resource adjustments. This log will serve as an audit trail and facilitate performance analysis, bottleneck identification, and future improvements.

Recruiter Agent (RA)

The RA is responsible for managing the full lifecycle of specialized agent involvement—from agent selection to collaborative task execution. Based on the complexity level determined by the CCA, the RA dynamically assigns and operationalizes the appropriate LLM-based agents. Beyond initial selection, the RA also orchestrates collaborative reasoning sessions to ensure task alignment and effective agent communication. Its core responsibilities include:

- *Agent Recruitment*: The RA maintains a registry of available agent capabilities and uses complexity tags (low, moderate, high) to match task requirements with agent expertise. For instance, a low-complexity task may trigger a single-agent response, while higher complexity levels result in composite agent group formation.
- *Dynamic Agent Composition*: In scenarios requiring cross-domain reasoning (e.g., security and QoS), the RA dynamically instantiates a multi-agent discussion panel tailored to the task at hand. This ensures domain-specific specialization is leveraged in a scalable and efficient manner.
- *Collaboration Orchestration*: For multi-agent tasks, the RA manages the interaction flow between agents by initiating structured reasoning sessions, coordinating turn-taking, and ensuring context is preserved across communication rounds.
- *Logging and Review Preparation*: The RA maintains a detailed record of agent interactions and responses, which is later reviewed by the decision-maker agent (DMA) to derive the final system-level verdict.

Domain-Specialized LLM Agents

Domain-specialized LLM agents are tailored for specific network issues and can address assigned tasks autonomously or collaborate with other agents.

1) Traffic Analysis Agent (TAA)

TAA addresses traffic-related issues such as congestion and packet loss. It analyzes the traffic data from various sources such as NetFlow and SNMP and identifies potential issues such as bottlenecks, traffic surges, and inconsistent flow patterns. For example, suppose the RA assigns a traffic-related task to TAA for resolution. In that case, it will take corrective measures such as adjusting network parameters, leveraging load balancing strategies, or rerouting traffic through alternative paths. For network issues of higher complexity (such as moderate or high), TAA participates in multi-round inter-agent discussions, where it exchanges analysis with relevant agents such as QAA and TDA to determine a jointly optimized mitigation strategy.

2) Quality Assurance Agent (QAA)

The QAA is dedicated to ensuring that the network maintains high service quality, particularly during peak demand periods or when resources are limited. It actively monitors performance metrics such as latency, jitter, packet loss, and bandwidth utilization. Leveraging these metrics, it identifies areas where service quality may fall below acceptable thresholds. For example, when latency spikes or bandwidth congestion occurs, the QAA adjusts allocation priorities, dedicating more bandwidth to high-priority applications requiring real-time responsiveness (e.g., VoIP or video streaming) while rate-limiting lower-priority flows. In more complex scenarios, such as multi-application congestion, the QAA engages in structured multi-round dialogue with other agents (i.e., TAA and TDA) to collaboratively resolve performance issues.

3) Threat Detection Agent (TDA)

The TDA consists of an LLM specialized in anomaly detection, and it is responsible for protecting the network against potential threats by continuously monitoring suspicious activity and implementing security protocols. In addition, it can evaluate the incoming security logs using different techniques and DL models for intrusion detection. These models are trained to detect patterns and identify potential security threats by analyzing large volumes of log data in real-time. TDA processes data from IDS, firewall logs, and other security feeds to identify unusual patterns or potential threats, such as increased failed login attempts, unusual data flows, or indicators of DDoS attacks. Upon detecting a threat, the TDA initiates appropriate countermeasures, such as blocking malicious IP addresses, restricting specific network paths, or escalating alerts for further investigation. TDA collaborates with other agents for complex network issues, such as coordinated attacks or advanced persistent threats. All actions of domain-specific agents are logged and summarized for review by the MAS coordinator and the decision-maker agent.

Decision Maker Agent (DMA)

DMA is responsible for synthesizing the outputs of recruited LLM agents to produce a final, context-aware resolution strategy tailored to the complexity level assessed by the CCA. In addition to reasoning over multi-agent insights, the DMA also initiates appropriate actions to implement the chosen solution. Its behavior is adaptive across varying levels of complexity:

- *Low Complexity:* For routine network issues—such as minor packet loss or routing issues, the DMA directly adopts the resolution proposed by a single specialized agent and executes the corresponding action after review.
- *Moderate Complexity:* For issues that span multiple domains, such as concurrent traffic congestion and minor security anomalies, the DMA reviews interaction logs from collaborating agents (e.g., TAA and TDA), synthesizes their insights, and orchestrates a coordinated response that balances performance and security.
- *High Complexity:* In critical scenarios involving system-wide disruptions—such as large-scale DDoS attacks or complex congestion patterns, the DMA performs detailed synthesis of structured reports generated by multiple agents (e.g., TAA, QAA, TDA). Based on this multi-perspective analysis, it formulates and initiates a comprehensive, system-level mitigation plan.

Results and Discussions

Data Description

To evaluate the feasibility of the proposed LaMA-SON framework, we performed a proof-of-concept experiment using the CIC-BCCC-NRC-IoMT-2024 dataset [14]. Currently, no publicly available dataset captures the full range of SON functionalities, namely self-configuration, self-optimization, and self-healing, which poses a significant challenge for benchmarking intelligent and adaptive frameworks for SON. Therefore, we adopt network traffic classification as a representative proxy task to assess our multi-agent system's collaborative reasoning and coordination capabilities in a realistic and data-driven setting. Although this dataset is not specifically designed for SON management, it provides fine-grained flow-level network traffic features in a variety of IoMT events that make it a practical choice to simulate SON-like behavior. Specifically, it contains diverse features such as protocol types, flow duration, packet sizes, port activity patterns, and inter-arrival statistics that are conceptually relevant for core SON functionalities such as capacity planning, QoS monitoring, and anomaly detection. Such semantic alignment of features enables an approximate yet informative evaluation of our multi-agent framework. In addition, to simulate domain-specific role alignment, task decomposition, and collaborative reasoning, we assign each agent with specific feature subsets that align with its intended domain role in this proxy setting.

For instance, the CCA evaluates overall traffic intensity by analyzing features such as flow duration, total forward and backward packet counts, flow byte rate, and inter-arrival times (IATs) statistics (including mean, minimum, and standard deviation). These features can help distinguish lightweight from complex sessions. The TAA is tasked to specifically focus on volume-related features such as forward/backward packet counts, packet lengths, packet rates, flow IAT metrics, and TCP (PSH/URG) flags. Similarly, QAA analyzes indicators of service quality, including forward and backward IATs, packet rates, active/idle durations, flow throughput, and down/up ratio. These features enable QAA to detect signs of congestion, delay, and instability that affect QoS. Finally, TDA focuses on features associated with malicious behavior that include protocol type, destination, and source IP and port features, window sizes, ACK flags, and connection attempts. These features allows TDA to detect port scans, unauthorized access attempts, and protocol misuse. This role-based feature assignment allows each agent to reason contextually while contributing to a coordinated multi-agent decision process.

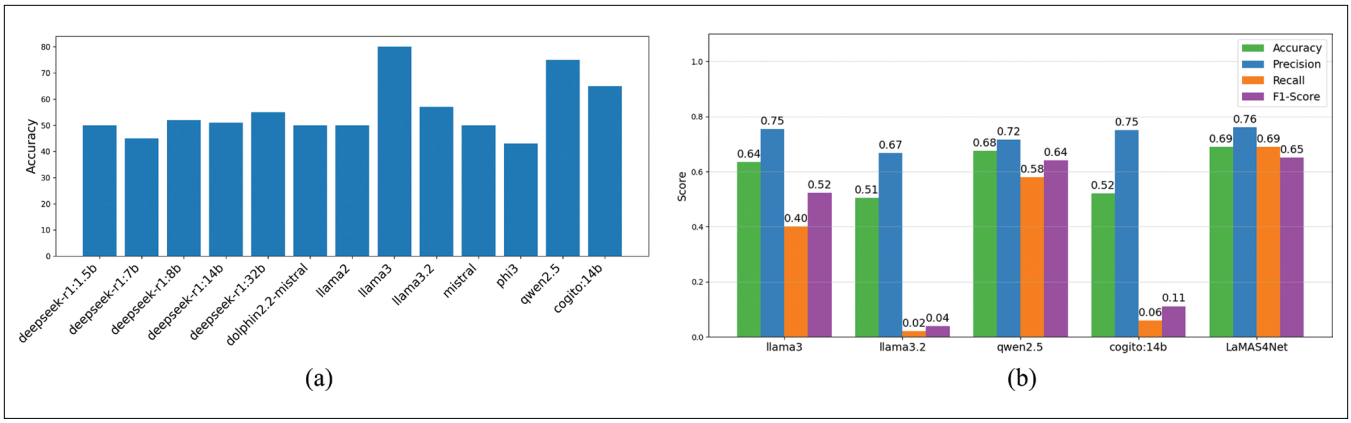


FIGURE 3. Model selection and performance evaluation for LaMA-SON: a) Zero-shot testing of 14 open-source LLMs on a lightweight test set to identify candidates for agent roles (based on performance), and b) Comparative analysis demonstrating the effectiveness of the LaMA-SON multi-agent architecture against top-performing baseline models.

Experimental Setup

To implement and evaluate the LaMA-SON, we developed a modular and agent-driven system using Python, leveraging the Camel-AI for multi-agent coordination and Ollama as the local inference engine for executing LLMs. Each agent within the LaMA-SON is instantiated as a ChatAgent via Camel-AI and invoked in isolated sessions to prevent contextual leakage across samples or review rounds. We reset the model state before every call and pass only the agent-specific feature view plus the current instruction. All LLMs run on the same host via the Ollama HTTP endpoint and are queried by agent keys. While Ollama does not natively support model parallelism on different ports, we ensure agent-level modularity through per-call isolation. We randomly sample 100 normal and 100 attack samples, resulting in a balanced subset of 200 traffic flow records for evaluation. Before inference, we remove non-essential features (IDs, IP addresses, labels) to avoid leakage, convert all time fields from microseconds to seconds, and add lightweight derived cues (e.g., IAT coefficient of variation, handshake/flag indicators, long-duration and low-rate flags). We used default temperature values for all models across agents except DMA, which has a temperature of 0.8. The multi-agent coordination was conducted for three rounds (initial predictions followed by two cyclic peer-review rounds), after each round the DMA issues the final decision.

Initial Model Screening

Before assembling the multi-agent architecture, we conducted a lightweight screening of 14 open-source LLMs using a minimalistic subset of 20 samples (10 normal, 10 attacks). Each model was evaluated under zero-shot settings using a unified prompt across the whole feature space to simulate baseline classification capability. As shown in Fig. 3(a), models such as `llama3.2`, `cogito:14b`, and `qwen2.5` demonstrated relatively high performance. They were subsequently selected for different agent roles within LaMA-SON based on both performance. Based on this empirical performance, we selected four models to serve as the backbone for the specialized agents in the LaMA-SON framework. The `llama 3.2` model was assigned to the CCA, while `qwen2. 5` powered the TAA. The QAA and the DMA were both supported by `llama3`, and the TDA was powered by `cogito: 14b`. Each agent receives a role-specific prompt, which is carefully designed to align its reasoning strictly with the subset of features relevant to its assigned function (i.e., role). For instance, TFA focuses on volumetric metrics (e.g., flow rate, IATs), while TDA emphasizes signature-based indicators (e.g., ports, flags, etc.), and QAA examines quality-related metrics (e.g., RTTs, congestion). Given the nuanced nature of the network traffic classification problem, the LaM-SON dynamic agent recruitment feature was not used in our evaluation.

Comparative Evaluation

To evaluate performance under more realistic conditions, we extended the experiment to the full 200-sample test set.

This evaluation compares our proposed LaMA-SON framework with four baseline models that performed best during initial screening: `llama3`, `llama3.2`, `qwen2.5`, and `cogito:14b`. Each baseline model is isolated using the same full-feature classifier prompt, whereas LaMA-SON utilized role-specific prompts, multi-agent feedback loops, and consensus-based reasoning. The comparative results are summarized in Fig. 3(b). In particular, all baseline models significantly degraded performance when scaled to the larger test set. For example, `llama3`'s accuracy dropped from 0.80 to 0.64, accompanied by a substantial decline in recall from 0.60 to 0.40. Similarly, the accuracy of `cogito: 14 b` model declined from 0.65 to 0.52, with a recall of only 0.06, indicating a high false negative rate. Such a drastic reduction in the performance of the baseline models highlights the sensitivity of standalone LLMs to increased data complexity and variance.

In contrast, LaMA-SON maintained stable performance with an accuracy of 0.69, precision of 0.76, recall of 0.69, and F1-score of 0.65. The notably higher recall achieved by LaMA-SON suggests that collaborative decision making improves the system's ability to detect malicious flows, a critical requirement in network intrusion detection, where false negatives can result in unmitigated attacks. We attribute this improvement to two key mechanisms: (1) feature-role

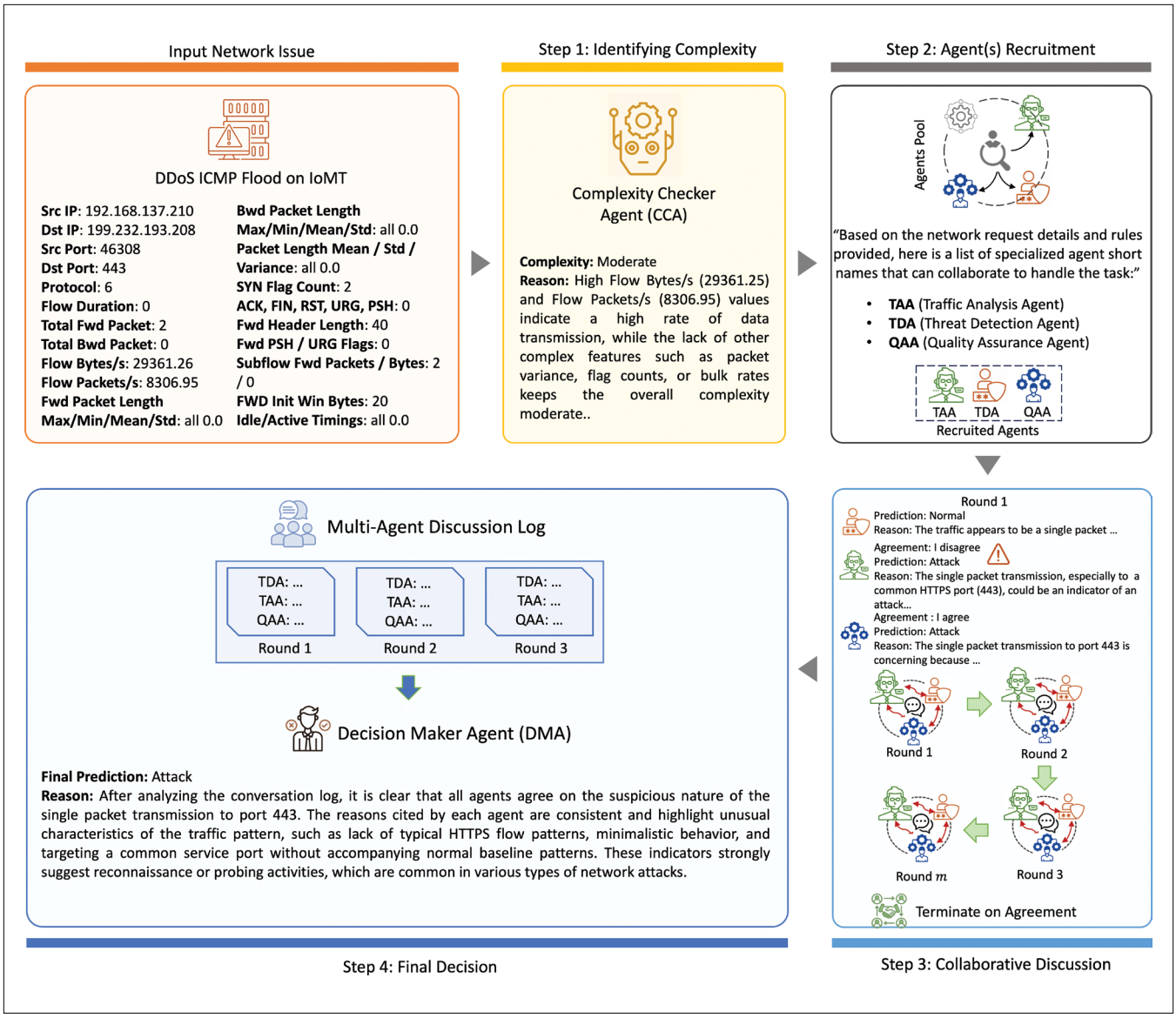


FIGURE 4. Actual execution of the LaMA-SON in analyzing a real moderate-complexity network flow from the CIC-BCCC-NRC-IoMT-2024 dataset. The workflow includes complexity assessment, agent recruitment, collaborative multi-round reasoning among LLM-based role-aligned agents, and a final decision synthesized by the DMA.

alignment, which constrains agents to reason only over task-relevant attributes, and (2) multi-agent dialogue, which enables corrective feedback loops during the reasoning process. Furthermore, our experiments revealed that standalone LLMs often misclassified sparse yet highly anomalous flows, such as isolated single-packet transmissions to privileged ports, as benign. These errors likely stemmed from the model's overreliance on dominant statistical patterns, ignoring semantic anomalies. In LaMA-SON, agents flagged such deviations independently in the review process (see Fig. 4, where TAA raised an early concern in the multi-round discussion phase). This ability to reason iteratively and reconcile conflicting responses is absent in the standalone LLMs.

The efficacy of LaMA-SON's collaborative reasoning paradigm, where inter-agent dialogue and role-specific analysis mitigate the limitations of isolated model inference, was consistently observed across multiple test flows. The consistent results also validate the advantage of structured role alignment and cross-verification in improving detection robustness across diverse traffic flows. To further illustrate LaMA-SON's collaborative inference capabilities, Fig. 4 illustrates an actual example of our evaluation, showing how LLM agents collaboratively analyze a moderate complexity network flow. After CCA classifies the flow as moderately complex, the system recruits specialized agents (TAA, QAA, and TDA) based on task-specific requirements. TDA initially produces an independent prediction, but rather than terminating at the first inference, LaMA-SON initiates a multi-round discussion process designed to resolve disagreement and refine predictions. In this process, the agents engage in three iterative rounds of discussion. During each round, they review the justifications offered by their peers and are given the opportunity to update their predictions (and responses) accordingly. This iterative mechanism is essential for getting complementary

insights (e.g., statistical anomalies, protocol mismatches, or heuristic cues) and ensuring that the final output reflects multi-perspective verification. Once consensus is reached, the DMA issues the final decision along with a rationale that transparently reflects the collaborative reasoning trail. This iterative discussion process directly addresses the limitations of standalone LLMs' inferences and provides context-aware decisions suitable for high-stakes network environments.

From ML/DL and LLMs to LLM Agents: Comparative Reflections

While traditional ML and DL models have long served as the backbone of automated decision-making in networking applications, their reliance on fixed architectures and task-specific training significantly limits adaptability in dynamic environments such as SONs. For instance, ML/DL models often struggle with generalization across unseen scenarios and lack mechanisms for contextual reasoning and collaborative adaptation. On the other hand, standalone LLMs offer substantial advancements in zero-shot reasoning and natural language understanding. However, they still operate in isolation and are inherently constrained by prompt design and limited task decomposition capabilities. Critically, standalone LLMs are often opaque in their decision logic and cannot natively support structured feedback, domain alignment, or multi-role inference. LaMA-SON addresses these limitations by employing LLMs within a MAS architecture that enables role specialization, collaborative inference, and transparent decision refinement. Through task decomposition and feature-role alignment, each agent in LaMA-SON operates over domain-relevant attributes, improving interpretability and domain fidelity. Multi-round dialogues between agents serve as a corrective mechanism, mitigating erroneous judgments that might arise in single-pass inference. Moreover, the modularity of LaMA-SON supports scalable deployment, where lightweight agents can be instantiated across base stations and synchronized through a cloud memory layer. This enables decentralized decision making and improves the resilience of the system, a critical requirement in modern SON environments. These advantages position LLM-based agent systems as a compelling alternative to conventional approaches, capable of bridging the gap between flexibility, domain-specific intelligence, and operational transparency. Table 2 summarizes the comparative advantages in key capabilities of traditional ML, DL, standalone LLMs, and multiple LLM agents.

Scalability, Evaluation, and Deployment Considerations

In this paper, we have presented a vision for deploying LLM-based MAS for operational management in SONs. Moreover, to demonstrate the practical feasibility of our approach, we performed an empirical evaluation in a simulated single-host environment using a proxy task. Furthermore, we would like to note that while LaMA-SON is not evaluated in distributed and multi-host settings, we envision that domain-specialized small LLMs (tailored for specific SONs tasks) can be deployed efficiently in a containerized manner across base stations. These agents can operate under lightweight orchestration frameworks (e.g., Docker, Kubernetes, etc.) with local execution to minimize latency, while maintaining global situational awareness through asynchronous cloud-shared memory.

Our current evaluation relies on prompt engineering and manual feature-role assignment to simulate agent behavior, and in our future work, we plan to develop a domain-specialized SON benchmark dataset and fine-tune small-scale LLMs that are tailored to the key SON tasks such as self-configuration, optimization, and fault management. In addition to benchmarking scalability and resource usage under distributed deployments,

SON Capability	Conventional ML/DL Models	Standalone LLMs	LaMA-SON (Collaborative LLM Agents)
Mobility Management	Task-specific models, require frequent retraining (lacks generalizability)	Generates textual responses but lacks KPI tracking	Multi-agent coordination enables adaptive parameter optimization based on temporal KPI trends
Load Balancing and Resource Allocation	Heuristics or supervised models per site or cluster (lacks generalizability)	Prompt-dependent responses without domain alignment	Agents collaboratively adjust resource weights across cells based on dynamic traffic insights
Anomaly Detection and Recovery	Anomaly classifiers trained per scenario, limited generalization	Can describe issues but lacks operational traceability	Role-specific agents detect, analyze, and resolve faults via structured diagnostic dialogue
Coordination Across SON Functions	Isolated pipelines, no inter-function communication	Monolithic inference, lacks modularity	Hierarchical LLM agents aligned to SON roles (e.g., traffic, QoS, security) collaborate iteratively
Tool Invocation	Requires custom logic or integrating middleware	Function-calling supported but not decomposed across operations	Agents can interpret intent and invoke domain-specific SON control functions autonomously
Adaptation to Unseen Network Conditions	Poor generalization, requires labeled retraining	Moderate adaptability via prompts, lacks memory or feedback loop	High adaptability via prompt grounding, inter-agent dialogue, and feedback-driven refinement
Interpretability and Reasoning	SHAP, LIME, or attention weights offer limited explainability	Natural language explanations, but no structured decision path	Agent reasoning steps are fully logged, enabling post-hoc traceability and auditability
Handling Multi-Factor Operational Scenarios	Performance degrades with increased variable complexity	Struggles without explicit scaffolding or decomposed input	Agents decompose complex tasks (e.g., QoS + security + traffic) across specialized roles

TABLE 2. Comparison of LaMA-SON with ML and DL Models and Standalone LLMs for SON Operations Management.

our future efforts will focus on incorporating task-specific and agent-centric metrics that can better reflect the performance of multi-agent systems. These include measures such as collaborative consistency (the agreement across agents in their inferences), agent conflict rate (the frequency of contradictory outputs), and redundancy ratio (overlap in reasoning across roles). Such metrics will provide deeper insights into the collective reasoning, coordination efficiency, and real-world viability of LLM-based MAS for intelligent networking.

Open Research Issues

While the proposed LaMA-SON framework demonstrates the promise of integrating LLM-based agents into complex networking tasks, several research challenges remain unaddressed. Below, we outline key open research issues that require further exploration to advance the deployment and reliability of LLM-based MAS in dynamic network environments.

1) Scalability of MAS

The integration of LLMs in MAS has attracted significant attention from the research community, and the potential of MAS has been demonstrated for various challenging applications in wireless networks. However, scaling LLM-based MAS to manage complex and large networks introduces challenges related to computational resources, communication efficiency, and system coordination. As the number of agents increases, communication bottlenecks and network congestion can arise, potentially degrading system performance. Therefore, addressing these challenges requires developing efficient LLM solutions to maintain performance in large-scale deployments.

2) Security and Privacy Issues in MAS

The architecture of a typical MAS poses significant security and privacy challenges. Therefore, ensuring secure communication between agents is crucial to prevent privacy breaches and unauthorized access. Similarly, it is essential to implement privacy-preserving mechanisms for shared memory to protect sensitive information exchanged across base stations or edge nodes. Moreover, the heterogeneous and distributed nature of MAS increases the attack surface, making them susceptible to various security threats. In particular, a significant concern is the identity management of participating agents in MAS. Without robust authentication, malicious entities may impersonate legitimate agents, leading to spoofing, data poisoning, or decision manipulation. Therefore, each agent must maintain a verifiable identity, which could be supported through cryptographic signatures, secure certificates, or decentralized identity frameworks such as blockchain-based identity registries. In addition, trust management models such as reputation-based scoring or behavioral consistency checks can be integrated for dynamically assessing the reliability of agents over time. These safeguards are essential to prevent agent masquerading, enforce role-based access control, and ensure trust across distributed coordination. For instance, Huang et al. [15] have already demonstrated the impact of malicious agents that can disrupt collaborative processes by generating incorrect or irrelevant outputs.

3) Long-Term Learning and Knowledge Sharing

In our LaMA-SON framework, we propose to establish a shared cloud memory to store global information and to facilitate collaboration between intelligent BS. This memory can be conceptualized as a centralized knowledge store (e.g., a vector database, graph-based memory, key-value system) that can be accessed by agents operating at different network layers. Such a memory module would support the asynchronous exchange of learned patterns (e.g., anomaly signatures, optimal configuration states), aggregated observations (e.g., traffic trends across BS), and context-rich information (e.g., policy constraints, environmental context, user mobility patterns) among agents. Moreover, to make this shared memory effective in dynamic network environments, efficient memory management and access strategies such as time-decayed relevance scoring, agent-specific embedding filters, or memory pruning heuristics can be used to ensure the system's adaptability, resilience, and long-term learning capacity.

4) Interoperability Issues

Integrating MAS into wireless communication networks presents significant interoperability challenges due to the diverse architectures and protocols involved. Therefore, it is crucial to develop universal frameworks to ensure seamless integration of MAS into legacy infrastructures, which requires careful consideration to balance customization and generalizability. A promising approach involves utilizing middleware solutions as intermediaries between MAS and existing systems. Middleware can facilitate communication and data exchange, enabling MAS to interact with legacy systems effectively.

5) Developing Specialized and Efficient LLMs

Deploying multiple LLMs at the BSs in SONs presents significant advantages in efficiently managing network operations. However, this integration increases costs since traditional LLMs demand substantial computational and memory resources. This hinders the large-scale deployment of multiple LLM agents on resource-constrained BSs. Developing specialized and TDAll LLMs tailored for BS environments is essential to address these challenges.

6) Lack of Task-specific Benchmarks

The lack of standardized evaluation frameworks for assessing LLMs-based MAS performance in real-world networks hinders the ability to measure adaptability, collaboration efficiency, and system resilience. Establishing comprehensive metrics and creating open-source benchmarks are necessary to facilitate consistent and transparent evaluation of MAS. Addressing these challenges is imperative for advancing LLM-based multi-agent systems in networking applications.

Conclusion

In this paper, we introduced **L**arge language model-based **M**ulti-**A**gent **S**ystem for **S**elf-**O**rganizing **N**etworks (LaMA-SON), which is designed to support adaptive and intelligent operations in SONs. The architecture targets three critical operational tasks: traffic management, quality of service (QoS) optimization, and security threat detection.

LaMA-SON integrates role-specialized LLM agents at the BS level, enabling decentralized decision-making, dynamic complexity handling, and collaborative resolution of network issues. To assess the feasibility of the proposed approach, we conducted a proof-of-concept evaluation using a real-world network traffic task in a proxy setting. In this setting, each domain-specialized agent operates on role-aligned features and engages in structured collaboration for traffic flow classification. Our results demonstrate that LaMA-SON achieves a balanced performance than standalone LLM baselines across various metrics. These findings validate the benefits of structured multi-agent collaboration for improving reasoning quality, reducing false negatives, and increasing decision resilience. Overall, LaMA-SON illustrates the potential of combining open-source LLMs with modular agent-based architectures for future SON deployments. Our work lays the foundation for more scalable, interpretable, and adaptive network management solutions that move beyond traditional static models. Future work includes extending this framework to domain-specific datasets while integrating dynamic agent recruitment and exploring memory-efficient LLM variants tailored for edge environments.

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